

Day 8 — Speed Sprints

Module: Multiplication Near a Power of 10 — Nikhilam · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: race traditional vs nikhilam on 30 problems.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *Ūrdhva-tiryagbhyāṁ* (IAST: Ūrdhva-tiryagbhyāṁ; lit. ““vertically and crosswise” — the general-purpose sūtra”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **Sprint A (Nikhilam-favourable, 10 problems, 4 min).** All factors within 5 of 100 or 1000.
2. **Sprint B (mixed, 10 problems, 4 min).** Mix of near-base and far-from-base.
3. **Sprint C (long-mult-favourable, 10 problems, 4 min).** All factors far from base.
4. **Analysis.** Each student records times. Most: $A < B < C$ dramatically. The CONCRETE evidence that Nikhilam works for ONE class.
5. **Discussion.** “*What’s your strategy choice now? When will you use Nikhilam in real life?*” Each student writes 2 sentences.

Activity (15 min)

A scaled version of *Method Speed Race* (see [activities/](#)). Today’s slice: focus on the part of the activity that reinforces Speed Sprints.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”

- Preview tomorrow.

Homework

(See `homework.md` for today's task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).